

Digital Research Toolkit for Linguists

Week 4: Looking at data

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Homework

Name your files with **!YOUR!** name and assignment number.

Type your answers in the comments **!NEXT!** to the word “ANSWER”.

What is the output of the following operations? **!If there is an error, what caused it?!**

→ A pattern you will see in your studies, work, experiments, teaching...

Homework

According to R, what is **the type** of the following variables?

→ Type, not output!

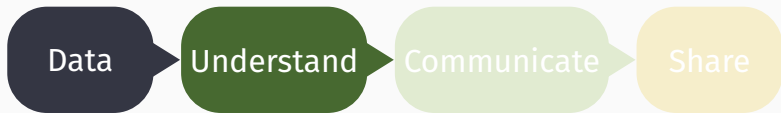
According to R, are the following two variables equivalent **(yes/no)**!

→ You only needed to run the existing code to get the answer.

Questions?

Table of contents

1. Cleaning and transforming data
2. Wrap-up



create, find,
data types,
formats,
encoding

R, packages,
import,
clean,
transform,
debug,
visualize,
export

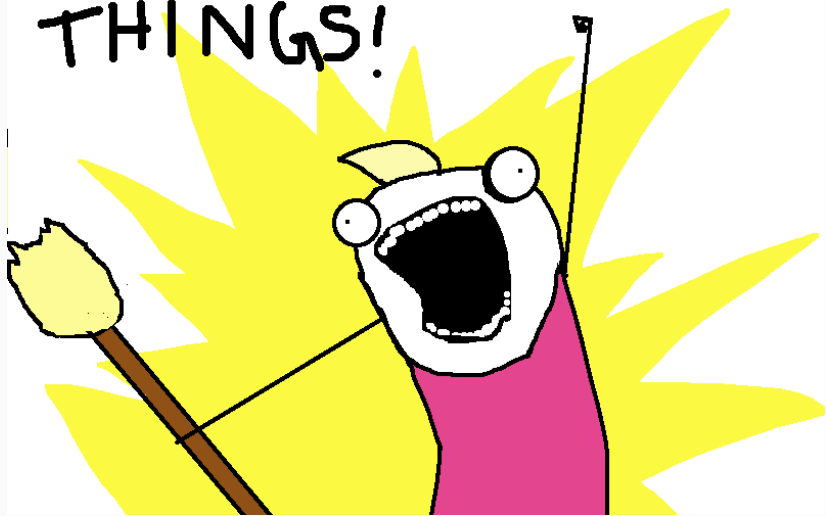
document,
create
clean and
beautiful
reports

connect,
collaborate,
backup,
replicate



Cleaning and transforming data

CLEAN ALL THE
THINGS!



Data cleaning steps

give descriptive names `rename( WHERE, NEW=OLD)`

select relevant columns `select( WHERE, WHAT)`

remove missing values `na.omit(WHERE)`

select values `filter( WHERE, TRUE CONDITION)`

reorder values `arrange(WHERE, HOW)`

create values `mutate( WHERE, NEW=FUNCTION(OLD))`

check unique values `unique( WHERE)`

Renaming

```
colnames(moses)  
rename(moses,  
       ID = MD5.hash.of.participant.s.IP.address,  
       ANSWER = Value)
```

What question does this experiment try to answer?

❓ What are the relevant columns?

👤 Participant ID: ID	Character
📄 Question ID: ITEM	Integer
📄 Question type: CONDITION	Integer
✎ Answer: ANSWER	Character
🕒 Response time: EventTime (kinda)	Double
📝 Right or wrong answer: ...	Logical

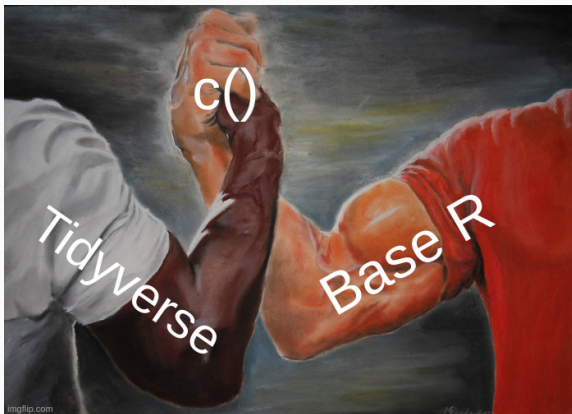
Selecting

Tidyverse

```
select(moses, ID, ITEM, CONDITION, ANSWER)  
select(moses, c(ID, ITEM, CONDITION, ANSWER))  
select(moses, c(ID, ITEM:ANSWER))
```

base R

```
moses$ID  
moses[ , "ID"]  
moses[ , c("ID", "ITEM", "CONDITION", "ANSWER")]  
moses[ , c(1,4:6)]  
subset(moses, subset = ITEM == 1, select = c(ID))
```



✓ `10 < 1`

✓ `print(10 < 1)`

✓ `c(10 < 1)`

✓ `cat(10 < 1)`

return/show

print

concatenate

concatenate & print

Task 1: Rename and drop columns

Look at the different columns in the `moses` data frame. Create a new data frame with the following changes:

1. Change the column name
`MD5.hash.of.participant.s.IP.address` to `ID` and
`Value` to `ANSWER`
2. Create a new data frame from the previous one with the
columns: `ITEM`, `CONDITION`, `ANSWER`, `ID`, `Label`,
`Parameter`

```
rename(WHERE, WHAT)
```

```
select(WHERE, WHAT)
```

Missing data

<code>na.omit(moses)</code>	everywhere
<code>na.omit(moses\$Item)</code>	only in the items
<code>na.omit(moses[, "Item"])</code>	only in the items
<code>na.omit(moses[, 4])</code>	only in the items

```
is.na(WHERE)  
is.na(select(moses, Item))
```


Task 2: Remove missing values

Create a new data frame from the previous one with no missing data anywhere.

```
na.omit(WHERE)
```

Coding basics: R as a calculator

addition	+
subtraction	-
division	/
multiplication	*
power	^
equals	==
not equals	!=
greater than	>
greater than or equal	>=
less than	<
less than or equal	<=
range	NUMBER1:NUMBER2
identify element	VALUE %in% OBJECT

negation

!

logical *and*

&

logical *or*

|

Sets



A



B



A & !B



!A & B



A | B



A & B



xor(A, B)

Filter (out)

<code>filter(moses, CONDITION == 1)</code>	condition 1
<code>filter(moses, CONDITION %in% 1)</code>	condition 1
<code>filter(moses, CONDITION >= 1 & CONDITION < 2)</code>	condition 1
<code>filter(moses, CONDITION == 1 CONDITION == 2)</code>	conditions 1-2
<code>filter(moses, CONDITION %in% 1:2)</code>	conditions 1-2
<code>filter(moses, CONDITION %in% c(1, 2))</code>	conditions 1-2
<code>filter(moses, CONDITION < 100)</code>	conditions 1-2
<code>filter(moses, CONDITION == 1:2)</code>	condition 1, if data is duplicated, then 1-2, don't do this

Task 3: Remove unnecessary rows

What are the relevant parts of the experiment?

❓ Do we need instructions, the exercise, final thank you screen, questions that are supposed to confuse you, ...

Create a new data frame from the previous one with rows that fulfill the following conditions. Keep the other columns.

1. `Parameter` is "Final"
2. `Label` is NOT "instructions"
3. `CONDITION` is only 1 or 2

```
filter(WHERE, TRUE CONDITION)
```

(Re)arrange

<code>arrange(moses, ITEM)</code>	item
<code>arrange(moses, ITEM, CONDITION)</code>	item, then condition
<code>arrange(moses, -ID)</code>	ID, descending
<code>arrange(moses, desc(is.na(ANSWER)))</code>	

Task 4: Sort the values

Create a new data frame from the previous one in which the values are sorted by **CONDITION** in ascending order, and **ANSWER** in descending order.

```
arrange(WHERE, HOW)
```


Re-code inconsistent information

<code>unique(VALUES)</code>	show all <i>unique</i> values
<code>unique(moses\$ANSWER)</code>	plain list
<code>unique(select(moses, ANSWER))</code>	only as many as fit on screen
<code>print(unique(select(moses, ANSWER)), n=Inf)</code>	show all

Task 5: Look at possible answers

Look at all answers give the the question with the item number 2.
Save all unique answers to a new variable.

```
select( WHERE, WHAT)  
filter( WHERE, TRUE CONDITION)  
unique( WHERE)  
+ assignment
```

Wrap-up

Summary

- ✓ renaming
- ✓ selecting
- ✓ dealing with missing data
- ✓ coding basics
- ✓ logic
- ✓ filtering
- ✓ arranging
- ▶▶ Pipes, mutating, grouping, summarizing, getting help, tidy code

Submit 1 R script.

- ❓ Complete logic exercise (next slide)
- ❓ Complete assignment 4 (→ ILIAS)
- ❗ Name your files with name and assignment number, e.g.
`pryslopskaassignment4.R`

Logic exercise

Your world has four individuals:



Octopus, dolphin, llama, and parrot.

Two are of the type **dive**, because they can dive:



Two are of the type **mammal**, because they're mammals:



Using only basic logical expressions (negation **!**, and **&**, or **|**) and the two group names, describe the sets on the next slide. Tips: a Venn diagram might help. You will have to use brackets for some answers.

  dive

  mammal

  !mammal

Homework assignment:

1. 
2. 
3. 
4. 
5.  
6.  
7.   
8.   
9.   
10.    
11. $\emptyset$ (exclude all)

Clean up on aisle “Answer”

```
cant_answer <- c("Can't Answer", "Can't answer",  
"Can't answer the question", "Can't answerer",  
"Can't be answered", "Can't answer", "i can't  
answer", "can't andwer" , "can't answer" , "can't  
answer (Nobel is given by Norway)", "can't asnwer",  
"can't know", "can't answer", "can't asnwer" ,  
"cant answer", "can't answ", "can't answer", "no  
answer")
```

i Try using `arrange()`, `filter()`, `select()`, and `unique()`.